

Response to JPE Replication Report

Subject: “Measuring Religion from Behavior: Violence, Climate Shocks and Religious Adherence in Afghanistan” (ID 20250296, Report iteration 1, June 26, 2026)

Thank you for the careful review of our replication package. We have addressed each point raised in the report, as described below. For clarity, we reproduce the report’s comments in black, indented and marked with a vertical bar, and we present our responses in blue. Comments are ordered as they appear in the report.

Report Section 2 (Summary)

1. Incomplete package: NUL-byte files and missing datasets

We obtained an incomplete package - there are several files which contain only a NUL byte character. This is typically the case if one does not properly sync an online-only dropbox folder - please verify that all files are present on your storage device before you create the zip of the replication package. As it stands, there was actually no dataset included in the package.

The 18 cleaning scripts, Code/Analysis/Fig1.R, and the data files all contained only NUL bytes because the package sat in a shared Dropbox folder and had not fully synced before it was zipped. This is the single cause of both problems the replicator encountered: the corrupted scripts and the absence of usable datasets.

We have since downloaded everything to local storage and verified that no file in the package is empty. All scripts contain valid code, and all data files are present and readable. We also re-ran the analysis scripts from the package and confirmed that they run and reproduce the tables in the paper.

2. Data provenance and status of the confidentiality agreement

In the parts we could check, we have some questions regarding data provenance. As you will know, all datasets used in the paper need to be cited (unless you generated them yourself). What is the status of your confidentiality agreement with the data provider?

All data sources are now cited in the README and in the paper; see our response to Item 6 for the full list.

Regarding the confidentiality agreement: the call detail records were provided to us by the telecommunications company pursuant to a signed confidentiality agreement that restricts our ability to communicate, disseminate, or disclose confidential information supplied by the company. The agreement provides continuing protection for information that is proprietary. Because the transaction-level CDR are proprietary, non-public telecommunications records supplied under this agreement, we are not authorized to include the underlying records in a publicly accessible replication archive. This is now stated as the first item in the list of confidential data in Section 2 of README.md.

Report Section 5 (Data description)

3. Preservation of restricted-access data

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

We have added a subsection, “Data preservation, access, and replication support,” at the end of the Data Availability Statement in README.md. It records that the authors commit to preserving these data and the associated code for a period of no less than five years following publication of the paper.

4. Affirmation of support for replication checks

{REQUIRED} Please provide affirmation of support for replication checks. - As per the JPE policy, “In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication.”

The same README subsection affirms that the authors will provide reasonable assistance to requests for clarification and replication, sets out what we can and cannot say about access to the confidential sources, and names a point of contact for access inquiries.

5. Codebook for the confidential data

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

We have expanded codebook.md, which now documents every file in the package as well as the confidential sources. For the raw CDR and the antenna geolocation file, we want to emphasize one point that may have been the source of the request: the variables listed are the raw, original field names of the data as delivered to us by the phone company. They are not derived variables that we constructed.

The replicator also suggested including additional summary statistics for the raw confidential data. Because these statistics would characterize the underlying proprietary CDR beyond the information already reported in the paper, we have not included them. We have instead provided the original field names and field-level descriptions needed to verify that a researcher obtaining authorized access has obtained substantially similar source data.

Report Section 5.1 (Data Sources)

6. Missing data citations

[Paraphrased] [From the Data Sources table:] The report marks “no data citation” (in the README and/or the paper) for: the transaction-level CDR, the individual-quarter CDR panel, the household survey microdata, the cell tower coordinates, the MODIS EVI data,

the village-level language/ethnicity data, the AIMS district boundaries, and the GADM country boundary.

We have added a citation for each, both in Section 9 (Data Citations) of README.md and at the corresponding place in the paper:

- Raw CDR and household survey microdata: these are now cited as data citations in the README and at the points in the paper where each source is first described. The citations are:

Anonymous Afghan Firm. n.d. "Call detail records covering 2013–2020." Unpublished data. Accessed June 29, 2026.

Blumenstock, Joshua, Michael Callen, Tarek Ghani, and Jacob Shapiro. 2016. "Household Survey Data on Households in Afghanistan." University of California, Berkeley. Confidential microdata.

- MODIS MOD13A3 EVI: previously cited only through the method we follow (Asher and Novosad 2020); the dataset itself is now cited. The citation is:

Didan, Kamel. 2021. "MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061." NASA Land Processes Distributed Active Archive Center. <https://doi.org/10.5067/MODIS/MOD13A3.061>

- SPEI inputs: we now cite the underlying raw data from which the SPEI is built — the ERA5-Land reanalysis data and the GMTED2010 elevation data. The citations are:

Muñoz-Sabater, Joaquín. 2019a. "ERA5-Land Hourly Data from 1981 to Present." Copernicus Climate Change Service (C3S) Climate Data Store (CDS). <https://doi.org/10.24381/cds.e2161bac>

Muñoz-Sabater, Joaquín. 2019b. "ERA5-Land Monthly Averaged Data from 1981 to Present." Copernicus Climate Change Service (C3S) Climate Data Store (CDS). <https://doi.org/10.24381/cds.68d2bb30>

Danielson, Jeffrey J., and Dean B. Gesch. 2011. "Global Multi-Resolution Terrain Elevation Data 2010 (GMTED2010)." U.S. Geological Survey Open-File Report 2011–1073. <https://pubs.usgs.gov/of/2011/1073/>

- Village-level language and ethnicity data: we previously named the source; it is now formally cited as a data citation. The citation is:

Afghanistan Information Management Service, Central Statistics Organization, United States Agency for International Development, and Yale University. 2012. "Afghanistan Village-Level Ethnicity Data (Primary Language)." Unpublished data. Accessed May 15, 2026.

- Geographic files: the AIMS 398-district boundary shapefile and the GADM country boundary shapefile are now cited in the README and in the paper's reference list. The citations are:

Empirical Studies of Conflict (ESOC). n.d. "Afghanistan Administrative Boundaries: 398 Districts." Princeton University. <https://esoc.princeton.edu/data/administrative-boundaries-398-districts>

Global Administrative Areas (GADM). 2018. "Global Administrative Areas, Version 3.6." <https://gadm.org/>

7. Codebook coverage of confidential sources, shapefiles, and raster

Shortcomings: the original CDR, individual-quarter CDR panel, household survey microdata, and cell tower coordinates are not available to the replicator; only synthetic stand-ins or aggregate derivatives are included. The codebook documents distributed files and derived variables, but not enough original-source detail to verify the confidential data if access were later granted. The package also does not provide a codebook entry for the included shapefiles or rasterwithmask.nc. No datasets are provided, including public ones.

On codebook coverage of the confidential sources, please see our response to Item 5: codebook.md now documents the confidential sources using the raw, original field names as delivered by the phone company.

The codebook also now gives field-level descriptions for the country boundary shapefile, the district boundary shapefile, and the grid-cell raster (rasterwithmask.nc).

On the absence of datasets from the package, please see our response to Item 1.

Report Section 6 (Requirements)

8. Deposit title

{REQUIRED} Please review the title of the deposit as per our guidelines. [Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)]

The README title now follows the required format: "Data and Code for: Measuring Religion from Behavior: Violence, Climate Shocks and Religious Adherence in Afghanistan." The deposit metadata will be set identically when the deposit is created.

9. Deposit authors and affiliations

{REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

The README now lists the authors with affiliations, in the same order as the manuscript: Oeindrila Dube (University of Chicago, NBER), Joshua E. Blumenstock (U.C. Berkeley), and Michael Callen (London School of Economics). The deposit metadata will match.

Report Section 7.1 (Potential Personal Identifiable Information)

10. PII flags

We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed

you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check! [Summary: Data files with PII indicators: 2 - Variables flagged in data: 6 - Code files with PII references: 12 - PII references in code: 519]

We reviewed all flagged references across the files, and confirm that none of them constitutes personally identifiable information; all are false positives. The flags are variable names in the econometric code, arguments to plotting functions, or substrings within longer tokens. For example, lat and lon are grid-cell centroid coordinates at roughly 10 km resolution rather than individual or tower locations; phone refers to the CDR transaction type, not to phone numbers; and son appears only inside the variable name season. The two flagged data files, grid_level.csv and grid_level.dta, likewise contain lat and lon only as grid-cell centroids, which are area-level identifiers that cannot be linked to individuals. All subscriber identifiers in our data are anonymized hashes.

Report Section 7.1.2 (File Paths Report)

11. Windows / mixed file paths

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for all STATA applications.) [Files Analyzed: 31; Windows Paths: 0; Unix Paths: 9; Mixed Paths: 2; Drive Letters: 0]

We reviewed the file paths in all 31 code files. Every path in the package is written with forward-slash separators and is portable across Stata, R, and Python on all supported platforms. We found no Windows-style separators and no drive letters in any path. The two instances flagged as mixed appear to be false positives. Only eight of the 31 files contain a backslash anywhere, and in each case the backslash belongs to a string that is not a path. We have additionally replaced the authors' machine-specific project-folder paths in the documentation-only cleaning scripts with the placeholder <PATH_TO_PROJECT_ROOT>; the absolute paths that remain in the cleaning scripts describe the secure server's internal directory structure and are retained for documentation.

Report Section 7.1.3 (Duplicate Files Report)

12. Duplicate files

[Paraphrased] We found the following duplicate files:
CDR_shortcode_raw_synthetic.csv and CDR_sms_raw_synthetic.csv (identical checksums).

The two files reported as duplicates were empty: like the scripts discussed in Item 1, they contained only NUL bytes as a result of the same Dropbox syncing issue. Both files are now present with their intended content.

Report Section 7.2 (Code description)

13. NUL-byte cleaning programs and Fig1.R

Eighteen of the 19 cleaning programs contain only NUL bytes and therefore cannot be reviewed. The only readable cleaning program is 13_precleaning_alltables.do. Fig1.R also contains only NUL bytes, so the code producing Figure 1 cannot be identified.

Please see our response to Item 1: these files were corrupted by the same Dropbox syncing issue and now contain valid, non-empty code.

14. No program for Figure A1

No program was provided for Figure A1.

Figure A1 (“Poster Prohibiting Cell Phone Use”) is a photograph of a poster rather than a figure generated from data, so no script was written for it. This is now noted in the list of figures in Section 5 of README.md.

15. No single controller for every output

There is no single controller for every output: Figure A2 and Table A13 are outside the figure and table controllers, respectively.

The package is run from four entry points rather than one master script because the outputs are produced in three different languages and no single program can call all of the others: All_Tables.do (Stata) is the controller for all tables except Table A13; All_Figures.R (R) is the controller for all figures except Figure A2; FigA2.qmd must be rendered with Quarto, so it cannot be called from the R figure controller; and TabA13.R is written in R, so it cannot be called from the Stata table controller. This is now documented in a “Note on entry points” in Section 4 of README.md.

16. In-text numbers not generated by identifiable code

Several substantive in-text numerical claims are not generated by identifiable code; these are listed separately in package-output-map.xlsx.

The in-text numerical claims listed in package-output-map.xlsx are the same three numbers flagged in Section 9.4 of the report. Each is addressed in our response to Item 21.

Report Section 8 (Replication)

17. Two zipped subfolders in the archive

I downloaded the code from GitHub and added the data from Dropbox, but there is an issue in the zipped replication package which contains two zipped subfolders: Fengzhe Liu - replication package and Fengzhe Liu - replication package_20260529. The first package contains the codebook and paper and appendix in pdf, I then assumed it was the correct one.

The archive contained two subfolders because we uploaded the package twice. The second folder was submitted after we were asked to include the paper and appendix in the folder. The replicator correctly identified the later folder as the definitive version. Our re-submission will contain a single top-level replication folder.

18. Package location must be set in three files

To generate output you need to update the subsequent files with the replication package location: - Code/Analysis/All_Tables.do - Code/Analysis/TabA13.R - FigA2.qmd

Each of the three files now sets the package root in a single, clearly marked placeholder, `<PATH_TO_REPLICATION_PACKAGE>`, which the replicator replaces with the local path to the package. Instructions appear in step 2 of Section 4 of README.md; all other R scripts resolve paths automatically and require no changes. The non-executable cleaning scripts use a separate placeholder, `<PATH_TO_PROJECT_ROOT>`, which refers to the authors' internal project folder rather than the replication package; see Section 4 of README.md.

Report Section 9 (Findings)

19. Not able to reproduce due to unreadable data files

Not able to reproduce code due to bad formatting of databases or total absence of data. The replication for Stata was impossible due to the bad formatting of the dta files. It may be due to the facts that stata does not support empty datasets. [Similar errors for the R code and the Quarto notebook.]

These errors were the consequence of the empty (NUL-byte) data files described in Item 1. All data files are now present and readable, and we have re-run the analysis scripts from the package and confirmed that they run and reproduce the tables and figures in the paper.

20. Table A6: duplicated panel letter E

There is a mistake in numbering in table A6, the letter E is duplicated.

The second panel E ("Variables at District-Year Level") was a typo and has been corrected to Panel F, in both the paper and All_Tables.do.

Report Section 9.4 (In-Text Numbers)

21. Three flagged in-text numbers

[Paraphrased] There are in-text numbers, but they are not identified in the code: (1) Paper PDF p. 2: a one-SD survey religiosity increase is associated with a 27% increase in the Maghrib dip. Later text and Table A2 report 29%. (2) Paper PDF p. 9: 90% of Afghans are Sunni - no package program. (3) Paper PDF p. 20 n. 23: adult literacy rate is 43% - no package program.

1. p. 2 (27% vs. 29%): As the replicator notes, the later text and Table A2 report 29%. The 29% figure is the correct one, and the text has been corrected to match it.

2. p. 9 (90% of Afghans are Sunni): This is background demographic context rather than a result computed from our data, which is why no program in the package generates it. The figure has been corrected to 89%, and a citation to the source has been added. On p. 9 we now write:

Much of our analysis focuses on the Maghrib ("sunset") prayer window, the fourth daily prayer window, which begins immediately after sunset. Under Sunni Islam, the

branch practiced by 89% of Afghans (U.S. Department of State, 2024), the Maghrib window opens precisely at sunset and must be completed by dusk.

The citation is:

U.S. Department of State. 2024. "2023 Report on International Religious Freedom: Afghanistan." https://www.state.gov/wp-content/uploads/2024/04/547499_AFGHANISTAN-2023-INTERNATIONAL-RELIGIOUS-FREEDOM-REPORT.pdf

3. p. 20, n. 23 (adult literacy rate is 43%): Likewise a contextual statistic not derived from our data. The figure has been corrected to 37%, and a citation to the source has been added. This passage is now footnote 24 on p. 21 of the revised draft, where we write:

These numbers are influenced by the country's low adult literacy rate (37%) (World Bank, 2021a) and limited smartphone penetration. Consequently, inferences made from text messages and data use would be less representative of the general Afghan population.

The citation is:

World Bank. 2021a. "Literacy Rate, Adult Total (% of People Ages 15 and Above) – Afghanistan." World Development Indicators. <https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=AF>